Product Requirements Document (PRD): AI-Powered Road Accident Detection and Emergency Alert System

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Executive Summary

AccidentGuard AI is an artificial intelligence-powered road accident detection and emergency alert system that automatically analyzes CCTV video footage to identify traffic accidents and instantly notify emergency responders via SMS and email. The system leverages state-of-the-art deep learning (YOLOv8) combined with GPS location tagging to provide comprehensive emergency response capabilities for road safety management.

Mission: Save lives by dramatically reducing emergency response times from hours to seconds through automated accident detection.

Target Market: Government agencies, city administrations, emergency response departments, and traffic management authorities in developing economies, with initial focus on Cameroon.

Key Value Proposition: Deploy life-saving accident detection using existing CCTV infrastructure without requiring expensive new hardware or vehicle-mounted sensors.

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1. Product Overview

1.1 Product Vision

To create the world's most accessible and effective automated accident detection system, specifically designed for low- and middle-income countries where road traffic fatalities are highest and emergency infrastructure is most limited.

1.2 Problem Statement

Core Problem: Road traffic accidents in Cameroon and similar developing economies suffer from critical detection and response delays:

- **6,000 annual deaths** in Cameroon from traffic accidents
- **2-4 hour response times** in rural areas; 15-45 minutes in urban centers
- **30% of accident deaths** attributable to delayed medical response
- **Existing CCTV infrastructure underutilized** for real-time safety monitoring

- **No automated detection mechanism** to instantly alert emergency services

Customer Pain Points:

1. **Emergency Services:** Cannot respond to accidents they don't know about; manual reporting is slow and unreliable
2. **Accident Victims:** Critical "golden hour" wasted waiting for someone to report incident
3. **City Administrators:** Cannot justify CCTV investments when cameras only serve post-incident investigation
4. **Government:** Unable to meet UN SDG targets for 50% reduction in road deaths by 2030

1.3 Solution Overview

AccidentGuard AI addresses these challenges through:

1. **Automated Video Analysis:** AI continuously monitors CCTV footage to detect accident events with 92.8% precision and 89.6% recall
2. **Instant Location Tagging:** GPS coordinates automatically assigned to every detected accident for emergency dispatch
3. **Multi-Channel Alerts:** SMS and email notifications delivered to responders within 0.5 seconds of detection
4. **Existing Infrastructure Leverage:** Works with offline video or live streams from existing CCTV cameras
5. **Low-Resource Optimization:** Operates on minimal hardware (CPU-only mode available) with no streaming infrastructure required

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2. Target Users and Stakeholders

2.1 Primary Users

User Persona 1: Emergency Dispatch Operator

- **Role:** Receives accident alerts and coordinates emergency response
- **Goals:** Get accurate, timely accident information with location details
- **Pain Points:** Manual reports delayed, incomplete, or inaccurate
- **How AccidentGuard Helps:** Automatic alerts with GPS coordinates, confidence scores, and timestamps enable immediate dispatch
- **Key Features:** SMS/email alerts, location mapping, incident details

User Persona 2: Traffic Management System Administrator

- **Role:** Manages CCTV infrastructure and monitors traffic safety
- **Goals:** Maximize utility of existing CCTV investments; improve road safety outcomes
- **Pain Points:** CCTV footage reviewed only after incidents; no real-time analysis
- **How AccidentGuard Helps:** Converts passive surveillance into active safety system
- **Key Features:** Web dashboard, video upload interface, detection configuration, performance analytics

User Persona 3: City/Government Safety Official

- **Role:** Oversees public safety initiatives and resource allocation

- **Goals:** Reduce traffic fatalities; justify infrastructure investments; meet policy targets
- **Pain Points:** Lack of data-driven insights; uncertain ROI on technology investments
- **How AccidentGuard Helps:** Quantifiable impact metrics (lives saved, response time reduction); cost-effective deployment
- **Key Features:** Performance reports, accident hotspot analysis, system analytics

2.2 Secondary Stakeholders

Medical Emergency Responders (Ambulance Services)

- Benefit from faster notification and precise location information
- Integration requirements: SMS/email compatibility with dispatch systems

Police and Fire Services

- Coordinated multi-agency response enabled by simultaneous alerts
- Integration requirements: Standard communication protocols

Insurance Companies

- Faster claim processing with documented incident timestamps
- Reduced fraud through video evidence
- Integration opportunity: API for incident data access

Urban Planners and Transportation Engineers

- Accident pattern data for infrastructure improvements
- Evidence-based road design decisions
- Integration opportunity: Data export for GIS analysis

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3. Core Features and Requirements

3.1 Feature Priority Framework

Must-Have (P0) - MVP Features:

- Video upload and preprocessing
- YOLOv8-based accident detection
- GPS location assignment
- SMS alert delivery
- Email alert delivery
- Basic web dashboard
- Detection confidence scoring
- Alert logging and audit trail

Should-Have (P1) - Version 1.1:

- Live video stream processing
- Multi-camera support (5-10 cameras)
- Real GPS integration (replacing simulation)
- Mobile-responsive interface
- Performance analytics dashboard

- Configurable alert thresholds

Nice-to-Have (P2) - Version 2.0:

- Mobile app for responders
- Incident severity classification
- Victim count estimation
- Multi-language support
- Audio analysis (crash sound detection)
- Weather data integration
- Predictive accident risk modeling

3.2 Functional Requirements

FR1: Video Processing Module

FR1.1: Video Upload

- System shall accept video files in MP4, AVI, MOV formats
- Maximum file size: 500 MB per upload
- Drag-and-drop interface support
- Progress indicator during upload
- Validation: Reject invalid formats with clear error messages

FR1.2: Video Preprocessing

- Extract frames at original frame rate (15-30 FPS)
- Resize frames to 640×640 pixels (YOLO standard)
- Normalize pixel values to [0,1] range
- Maintain metadata: timestamp, camera ID, location
- Performance target: Process 30 FPS on GPU, 5 FPS on CPU

FR1.3: Frame Analysis

- Sequential frame-by-frame processing
- Support for batch processing of multiple videos
- Real-time status updates during processing
- Error recovery: Continue processing if individual frames fail

FR2: Accident Detection Module

FR2.1: Object Detection

- YOLOv8 model for vehicle and pedestrian detection
- Detect classes: car, truck, bus, motorcycle, bicycle, person
- Confidence threshold: 0.5 (configurable 0.3-0.9)
- Bounding box output with class labels and confidence scores
- Performance: 95%+ accuracy on vehicle detection

FR2.2: Multi-Object Tracking

- DeepSORT algorithm for tracking objects across frames
- Assign unique track IDs to each detected object
- Maintain trajectory history (position, velocity, direction)
- Track lifecycle management (creation, maintenance, termination)

- Handle occlusions and re-identifications

FR2.3: Anomaly Detection and Collision Classification

- Analyze trajectories for anomalous patterns:
  - Sudden deceleration: $\Delta v > 5$ m/s²
  - Proximity violations: Distance < safe threshold (vehicle-dependent)
  - Sudden direction changes: $|\Delta\theta| > 30°$
  - Vehicles leaving lane boundaries
- Multi-frame confirmation (3-5 consecutive frames) to reduce false positives
- Accident confidence scoring: 0.0-1.0 scale
- Alert threshold: 0.9 (90% confidence minimum)

FR2.4: Detection Accuracy Requirements

- Precision: ≥90% (minimize false alarms)
- Recall: ≥85% (maximize accident detection)
- F1-Score: ≥0.87 (balanced performance)
- Day/night performance ratio: ≥80% (night recall vs. day recall)
- Processing latency: <100ms per frame

FR3: GPS and Location Module

FR3.1: Camera Registration

- Admin interface to register camera locations
- Input fields: Camera ID, Name, Latitude, Longitude, Description
- Validation: GPS coordinates within valid ranges (-90 to 90 lat, -180 to 180 lon)
- Storage: Persistent database of camera locations
- Edit/delete capabilities for registered cameras

FR3.2: Location Assignment

- Automatic GPS coordinate assignment to detected accidents
- Link accident to camera location (±100m accuracy acceptable)
- Include location in alert messages (latitude, longitude format)
- Support for location visualization on maps (Google Maps integration)
- Fallback: Use approximate location if camera not registered

FR3.3: Future: Real GPS Integration

- API integration with vehicle tracking systems (deferred to v2.0)
- Support for dynamic location updates
- GPS accuracy validation and error handling

FR4: Alert Generation and Delivery Module

FR4.1: Alert Message Formatting

- Generate concise alert messages including:
  - Accident timestamp (ISO 8601 format)
  - GPS coordinates (decimal degrees)
  - Confidence score (percentage)
  - Camera ID and name

    - Incident type (collision, rollover, etc.)
- SMS limit: 160 characters (single segment)
- Email: Full details with optional video frame attachment

FR4.2: SMS Alert Delivery

- Integration with Twilio SMS API
- Support for multiple recipient phone numbers (5-20 contacts)
- Delivery confirmation tracking
- Retry logic: 3 attempts with exponential backoff
- Target delivery time: <5 seconds from detection
- Success rate: >99%

FR4.3: Email Alert Delivery

- Integration with Twilio SendGrid or SMTP
- Support for multiple recipient email addresses
- HTML email formatting with structured information
- Optional attachment: Accident frame image
- Delivery confirmation tracking
- Target delivery time: <10 seconds

FR4.4: Alert Logging and Audit Trail

- Log all alerts sent to database
- Record: timestamp, recipients, delivery status, incident details
- Searchable alert history
- Export capability (CSV, JSON)
- Retention: Minimum 1 year of alert history

FR5: User Interface and Dashboard

FR5.1: Video Upload Interface

- Drag-and-drop upload area
- File browser selection option
- Upload progress bar with percentage
- Preview uploaded video metadata
- Cancel upload capability

FR5.2: Real-Time Monitoring Dashboard

- Live video playback during processing
- Overlay bounding boxes on detected objects
- Display track IDs and trajectories
- Highlight detected accidents (red bounding box)
- Show confidence scores for detections
- Frame-by-frame navigation controls
- Pause/resume processing

FR5.3: Configuration Panel

- Camera location management (add/edit/delete)
- Responder contact management (add/edit/delete)

- Detection threshold configuration (sliders 0.5-0.95)
- Alert settings (enable/disable SMS, email)
- System parameter tuning (advanced mode)

FR5.4: Results and Analytics Dashboard

- Accident detection history (table view)
- Performance metrics display:
    - Total videos processed
    - Accidents detected
    - Alerts sent
    - Success/failure rates
- Filtering by date range, camera, confidence
- Export results to CSV
- Visual charts: Detection trends, accuracy metrics

FR5.5: Reports Section

- System performance summary
- Detection accuracy metrics (precision, recall, F1)
- Alert delivery statistics
- Resource utilization (processing time, memory)
- Accident hotspot map visualization
- PDF report generation

FR6: System Administration

FR6.1: User Authentication (Future)

- Login system with username/password
- Role-based access control (admin, operator, viewer)
- Password reset functionality
- Session management

FR6.2: System Configuration

- Model selection (YOLOv8 variants: nano, small, medium)
- Performance mode (CPU-only, GPU-accelerated)
- Processing queue management
- Database backup and restore

FR6.3: System Health Monitoring

- Uptime tracking
- Error logging and alerting
- Resource utilization metrics (CPU, memory, disk)
- Service status indicators (API connectivity, model loading)

3.3 Non-Functional Requirements

NFR1: Performance

- **Frame Processing Rate:** ≥25 FPS on GPU (NVIDIA RTX 2060 or equivalent); ≥5 FPS on CPU (Intel i7 or equivalent)
- **Detection Latency:** <100ms per frame (inference + tracking + analysis)
- **Alert Delivery Time:** <5 seconds from accident detection to SMS/email sent
- **System Response Time:** Web interface interactions complete within 2 seconds
- **Concurrent Users:** Support 5 simultaneous dashboard users
- **Video Processing Queue:** Handle 10 concurrent video processing jobs

NFR2: Reliability

- **System Uptime:** 99.5% availability (excluding planned maintenance)
- **Detection Accuracy:** ≥90% precision, ≥85% recall maintained across diverse conditions
- **Alert Delivery Success Rate:** >99% SMS/email delivery confirmation
- **Error Recovery:** Graceful handling of network failures, API errors, invalid inputs
- **Data Integrity:** Zero data loss for logged alerts and system configurations
- **Fault Tolerance:** Automatic restart on system crashes with alert notification

NFR3: Scalability

- **Multi-Camera Support:** Architecture supports 10-100 cameras (version 1.1)
- **Video Archive:** Support 1TB+ video storage
- **Database Growth:** Handle 1M+ detection records without performance degradation
- **Network Bandwidth:** Operate effectively with 5 Mbps internet connection (offline mode)
- **Cloud Deployment:** Support horizontal scaling via containerization (Docker, Kubernetes)

NFR4: Security

- **Data Encryption:** TLS/SSL for API communications; encrypted database storage
- **Access Control:** Role-based permissions for system administration
- **API Security:** Secure credential storage for Twilio, external services
- **Privacy Compliance:** Anonymization of stored video data; GDPR/local law compliance
- **Audit Logging:** Comprehensive logs of system access, configuration changes, alerts
- **Secure Deployment:** Follow OWASP security best practices

NFR5: Usability

- **Interface Intuitiveness:** System usable by operators with 1-day training
- **Error Messages:** Clear, actionable error messages for users
- **Help Documentation:** Comprehensive user guide and tooltips
- **Language Support:** English (MVP); French planned for Cameroon deployment
- **Accessibility:** WCAG 2.1 Level AA compliance for web interface
- **Mobile Compatibility:** Responsive design for tablet/mobile viewing (version 1.1)

NFR6: Maintainability

- **Code Quality:** PEP 8 Python standards; 80%+ test coverage
- **Documentation:** Complete API documentation; inline code comments
- **Modular Architecture:** Clear separation of concerns (detection, tracking, alerts, UI)
- **Logging:** Comprehensive logging for debugging and performance analysis
- **Version Control:** Git repository with documented commit history
- **Deployment:** Automated deployment via scripts or CI/CD pipeline

NFR7: Compatibility

- **Browser Support:** Chrome 90+, Firefox 88+, Safari 14+, Edge 90+
- **Operating Systems:** Linux (Ubuntu 20.04+), Windows 10+, macOS 11+
- **Video Formats:** MP4 (H.264/H.265), AVI, MOV
- **Database:** SQLite (development), PostgreSQL (production)
- **Cloud Platforms:** Replit, AWS, Google Cloud, Azure
- **APIs:** Twilio SMS, SendGrid Email, Google Maps (optional)

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4. Technical Architecture

4.1 System Architecture

Three-Tier Architecture:

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┌─────────────────────────────────────────────────────────────┐
│ PRESENTATION LAYER (Frontend) │
│ Streamlit Web Dashboard | Video Player | Reports UI │
└─────────────────────────────────────────────────────────────┘
↓
┌─────────────────────────────────────────────────────────────┐
│ APPLICATION LAYER (Backend) │
│ Video Processing | Detection Engine | Alert Manager │
│ Configuration Service | Analytics Engine │
└─────────────────────────────────────────────────────────────┘
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┌─────────────────────────────────────────────────────────────┐
│ DATA LAYER │
│ Video Storage | Model Weights | SQLite/PostgreSQL DB │
│ Alert Logs | Configuration Data │
└─────────────────────────────────────────────────────────────┘
↓
┌─────────────────────────────────────────────────────────────┐
│ EXTERNAL SERVICES │
│ Twilio SMS API | SendGrid Email | Google Maps API │
└─────────────────────────────────────────────────────────────┘
```

4.2 Technology Stack

Core Technologies:

- **Programming Language:** Python 3.8+
- **Deep Learning Framework:** PyTorch 2.0+ (YOLOv8 support)
- **Computer Vision:** OpenCV 4.7+
- **Object Tracking:** DeepSORT algorithm
- **Web Framework:** Streamlit 1.28+ (MVP); Flask/FastAPI (future)
- **Database:** SQLite (development), PostgreSQL 14+ (production)
- **Cloud Platform:** Replit (MVP), AWS/GCP (production)

Key Libraries:

- **Data Processing:** NumPy 1.20+, Pandas 1.3+
- **Machine Learning:** Scikit-learn 1.0+
- **Visualization:** Matplotlib 3.5+, Seaborn 0.11+
- **API Integration:** Requests, Twilio Python SDK
- **Video Processing:** FFmpeg (backend)

APIs and External Services:

- **Twilio SMS API:** Alert delivery to mobile phones
- **Twilio SendGrid:** Email alert delivery
- **Google Maps API:** Location visualization (optional)

Development Tools:

- **Version Control:** Git, GitHub
- **IDE:** VS Code, PyCharm, Jupyter Notebook
- **Testing:** pytest, unittest
- **CI/CD:** GitHub Actions (future)
- **Containerization:** Docker (future)

4.3 Data Flow

Video Processing Flow:
Video Upload → Frame Extraction → Preprocessing → YOLOv8 Detection → DeepSORT Tracking → Anomaly Analysis → Accident Classification → GPS Assignment → Alert Generation → SMS/Email Delivery → Logging

Alert Delivery Flow:
Accident Detected (confidence >0.9) → Retrieve Camera Location → Format Alert Message → Twilio API Call → Delivery Confirmation → Log to Database → Update Dashboard

4.4 Database Schema (Simplified)

Tables:

1. **cameras**
    - camera_id (PK)
    - camera_name
    - latitude

    - longitude
    - description
    - created_at
2. **responders**
    - responder_id (PK)
    - name
    - phone_number
    - email
    - role (police, ambulance, fire)
    - active (boolean)
3. **detections**
    - detection_id (PK)
    - video_id
    - camera_id (FK)
    - frame_number
    - timestamp
    - confidence_score
    - incident_type
    - latitude
    - longitude
    - created_at
4. **alerts**
    - alert_id (PK)
    - detection_id (FK)
    - recipient_type (sms, email)
    - recipient_contact
    - message_content
    - delivery_status
    - sent_at
    - delivered_at
5. **videos**
    - video_id (PK)
    - filename
    - upload_timestamp
    - processing_status
    - camera_id (FK)
    - total_frames
    - detections_count

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5. User Stories and Use Cases

5.1 User Stories

Epic 1: Accident Detection

- **US-001:** As a system operator, I want to upload CCTV video files so that the system can analyze them for accidents
- **US-002:** As a system administrator, I want the system to automatically detect vehicles and pedestrians in videos so that potential accidents can be identified

- **US-003:** As an emergency dispatcher, I want the system to accurately identify accident events with high confidence so that I only receive actionable alerts
- **US-004:** As a traffic safety officer, I want to see detection confidence scores so that I can assess alert reliability

Epic 2: Location and Context

- **US-005:** As a system administrator, I want to register camera locations with GPS coordinates so that accidents can be geolocated
- **US-006:** As an emergency responder, I want to receive GPS coordinates with every alert so that I can navigate to the incident location
- **US-007:** As a dispatcher, I want to see accident location on a map so that I can visualize incident geography

Epic 3: Emergency Alerts

- **US-008:** As an emergency dispatcher, I want to receive SMS alerts within seconds of accident detection so that I can dispatch responders immediately
- **US-009:** As a supervisor, I want email notifications with detailed incident information so that I can review and coordinate response
- **US-010:** As a system administrator, I want to configure alert recipient contacts so that notifications reach the right people
- **US-011:** As an operations manager, I want to see alert delivery confirmations so that I know responders were notified

Epic 4: System Monitoring and Analytics

- **US-012:** As a system administrator, I want to view real-time processing status so that I can monitor system operation
- **US-013:** As a traffic safety analyst, I want to view accident detection history so that I can identify patterns and hotspots
- **US-014:** As a government official, I want to see performance metrics (precision, recall, response times) so that I can assess system effectiveness
- **US-015:** As an administrator, I want to export detection results so that I can analyze data in external tools

5.2 Use Case Diagram

Actors:

- System Administrator
- Emergency Dispatcher
- Traffic Safety Officer
- System (Automated Agent)

Use Cases:

1. Upload Video
2. Process Video for Detection
3. Detect Accident
4. Generate Alert
5. Send SMS Notification
6. Send Email Notification

7. Register Camera Location
8. Manage Responder Contacts
9. View Detection Results
10. Configure System Parameters
11. Generate Performance Report
12. View Alert History

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6. Success Metrics and KPIs

6.1 Product Success Metrics

Detection Performance:

- **Precision:** ≥90% (minimize false alarms)
- **Recall:** ≥85% (maximize accident detection)
- **F1-Score:** ≥0.87 (balanced performance)
- **False Positive Rate:** <2% (maintain responder trust)

System Performance:

- **Alert Delivery Time:** <5 seconds (95th percentile)
- **Processing Throughput:** ≥25 FPS on GPU
- **System Uptime:** 99.5%+
- **Alert Delivery Success Rate:** >99%

User Satisfaction:

- **User Satisfaction Score:** ≥4.0/5.0
- **System Usability Scale (SUS):** ≥70/100
- **Training Time:** ≤1 day for operators
- **User Retention:** ≥80% after 3 months

Business Impact:

- **Response Time Reduction:** ≥90% vs. manual reporting
- **Lives Saved (Projected):** 150-1,000 annually in Cameroon
- **Cost per Life Saved:** <$50,000
- **ROI:** Positive within 2 years

6.2 Key Performance Indicators (KPIs)

Weekly KPIs:

- Videos processed
- Accidents detected
- Alerts sent (SMS + email)
- Average alert delivery time
- System uptime percentage
- False alarm count

Monthly KPIs:

- Detection accuracy (precision/recall/F1)

- User satisfaction score
- System error rate
- Processing capacity utilization
- Alert delivery success rate
- Active users count

Quarterly KPIs:

- Lives saved (estimated from response time reduction)
- Geographic coverage (cameras deployed)
- System scalability (cameras per instance)
- Cost per alert delivered
- Return on investment (economic benefit)

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7. Release Plan and Roadmap

7.1 Phase 1: MVP (Months 1-3)

Milestone 1.0 - Core Detection System

- Video upload and preprocessing
- YOLOv8 accident detection
- DeepSORT tracking
- Basic anomaly detection
- Simulated GPS location assignment
- SMS alert delivery (Twilio)
- Email alert delivery
- Basic Streamlit dashboard
- Alert logging

Deliverables:

- Functional prototype deployed on Replit
- User documentation
- System demonstration video
- Performance evaluation report

Success Criteria:

- Detection precision ≥90%, recall ≥85%
- Alert delivery <5 seconds
- System operational on cloud platform

7.2 Phase 2: Production Readiness (Months 4-6)

Milestone 1.1 - Enhanced Features

- Live video stream processing
- Multi-camera support (5-10 cameras)
- Real GPS integration
- Mobile-responsive UI
- Performance analytics dashboard
- Configurable detection thresholds

- User authentication and roles
- Database migration to PostgreSQL

Deliverables:

- Production-grade deployment
- Administrator training materials
- API documentation
- System architecture documentation

Success Criteria:

- Multi-camera coordination operational
- Live stream processing at 25 FPS
- User authentication working
- Production deployment stable

7.3 Phase 3: Advanced Capabilities (Months 7-12)

Milestone 2.0 - Intelligent Features

- Mobile app for responders
- Incident severity classification
- Victim count estimation
- Weather data integration
- Audio analysis (crash sound detection)
- Predictive accident risk modeling
- Integration with emergency dispatch systems
- Multi-language support (English, French)

Deliverables:

- Mobile application (iOS/Android)
- Advanced analytics module
- Integration APIs for third-party systems
- Comprehensive system evaluation

Success Criteria:

- Mobile app deployed to app stores
- Integration with actual dispatch systems
- Predictive modeling accuracy ≥80%
- Deployment in 2-3 Cameroon cities

7.4 Phase 4: Scale and Optimization (Year 2)

Milestone 3.0 - National Deployment

- Multi-city deployment (5+ cities)
- Support for 100+ cameras
- Real-time incident prediction
- Integration with national emergency services
- Advanced geospatial analytics
- Continuous model improvement pipeline

- Public API for third-party integrations

Deliverables:

- National-scale deployment
- Public developer documentation
- Research publications
- Impact assessment report

Success Criteria:

- 100+ cameras operational
- 1,000+ lives saved annually (projected)
- System sustainability demonstrated
- International interest and partnerships

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8. Risks and Mitigation

8.1 Technical Risks

Risk 1: Detection Accuracy Below Target

- **Probability:** Medium
- **Impact:** High (affects user trust and safety)
- **Mitigation:**
  - Use proven YOLOv8 architecture with strong baseline performance
  - Extensive testing on diverse datasets (CADP, local footage)
  - Iterative model fine-tuning based on real-world feedback
  - Conservative confidence thresholds (90%) to minimize false positives
- **Contingency:** Fallback to human-in-the-loop confirmation for borderline cases

Risk 2: Network/API Failures

- **Probability:** Medium
- **Impact:** High (alerts not delivered)
- **Mitigation:**
  - Redundant alert channels (SMS + email)
  - Retry logic with exponential backoff
  - Alert queue persistence to database
  - Monitoring and automatic restart mechanisms
- **Contingency:** Manual notification procedures for system operators

Risk 3: Insufficient Processing Performance

- **Probability:** Low
- **Impact:** Medium (slow processing, delayed alerts)
- **Mitigation:**
  - GPU optimization (CUDA acceleration)
  - Model quantization for edge deployment
  - Horizontal scaling via containerization
  - Performance profiling and optimization
- **Contingency:** Reduce frame rate or resolution; prioritize critical cameras

8.2 Operational Risks

Risk 4: Low User Adoption

- **Probability:** Medium
- **Impact:** High (system underutilized)
- **Mitigation:**
  - Early stakeholder engagement (government, emergency services)
  - Comprehensive training programs
  - User-friendly interface design
  - Clear value demonstration (pilot program results)
- **Contingency:** Phased rollout with intensive support; adjust based on feedback

Risk 5: Data Privacy and Legal Concerns

- **Probability:** Medium
- **Impact:** High (deployment blocked, legal liability)
- **Mitigation:**
  - Privacy-by-design architecture (anonymization, encryption)
  - Compliance review with legal experts
  - Transparent communication about data usage
  - Regulatory approval before full deployment
- **Contingency:** Adjust system design to meet regulatory requirements; engage legal counsel

Risk 6: Funding and Sustainability

- **Probability:** Medium
- **Impact:** High (project termination)
- **Mitigation:**
  - Demonstrate ROI with pilot results
  - Seek multiple funding sources (government, NGOs, insurance)
  - Low operational costs via cloud efficiency
  - Open-source model to attract community support
- **Contingency:** Scale back to core features; seek international partnerships

8.3 External Risks

Risk 7: Infrastructure Limitations (Power, Internet)

- **Probability:** High (in rural Cameroon)
- **Impact:** Medium (system unavailable in some areas)
- **Mitigation:**
  - Offline processing capability (no streaming required)
  - Low bandwidth requirements (SMS alerts)
  - Battery backup for critical installations
  - Progressive deployment (urban first, rural later)
- **Contingency:** Focus on urban corridors with reliable infrastructure initially

Risk 8: Competing Technologies or Solutions

- **Probability:** Low
- **Impact:** Medium (market competition)

- **Mitigation:**
  - First-mover advantage in Cameroon context
  - LMIC-optimized design (unique value proposition)
  - Open-source community building
  - Continuous innovation and improvement
- **Contingency:** Differentiate through superior accuracy, cost-effectiveness, or localization

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9. Dependencies and Assumptions

9.1 Dependencies

Technical Dependencies:

- Python 3.8+ runtime environment
- PyTorch/TensorFlow availability
- OpenCV library compatibility
- Twilio API service availability and affordability
- Internet connectivity for API calls (5 Mbps minimum)
- GPU availability for training and inference (optional for deployment)
- Cloud platform access (Replit, AWS, or similar)

Data Dependencies:

- Access to CADP dataset or equivalent for training
- Availability of local CCTV footage for testing
- Camera location data (GPS coordinates)
- Emergency responder contact information

Organizational Dependencies:

- Government approval for deployment
- Emergency services cooperation for alert integration
- CCTV infrastructure access
- Training resources for system operators
- Ongoing maintenance and support commitment

External Dependencies:

- Twilio service reliability (99.95% SLA)
- Email delivery infrastructure (SendGrid or SMTP)
- Internet service provider reliability
- Mobile network coverage for SMS delivery
- Regulatory approval for surveillance and data processing

9.2 Assumptions

Technical Assumptions:

- YOLOv8 performance transfers to Cameroon traffic scenarios
- Existing CCTV cameras provide sufficient resolution (≥720p)
- GPS accuracy of ±100 meters is acceptable for emergency response
- SMS remains reliable communication channel in Cameroon

- Python and open-source tools remain freely available

Operational Assumptions:

- Emergency services can integrate SMS/email alerts into dispatch workflows
- System operators trainable within 1 day
- CCTV footage accessible for processing (legal/technical)
- Responder contact information maintainable and up-to-date
- System maintenance resources available (IT staff, budget)

Business Assumptions:

- Government commitment to reducing road fatalities
- Stakeholder willingness to adopt automated systems
- Cost-benefit analysis demonstrates positive ROI
- Pilot program results sufficient to justify scaling
- International interest and partnerships feasible
- Funding available for initial deployment and operations

Environmental Assumptions:

- Internet connectivity available at deployment sites
- Power supply stable or backup systems feasible
- Weather conditions do not completely obstruct camera visibility
- Traffic patterns similar to training data scenarios
- Accident types in Cameroon similar to CADP dataset

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10. Success Criteria and Acceptance

10.1 MVP Acceptance Criteria

Functional Acceptance:

- [ ] System successfully uploads and processes MP4 video files
- [ ] YOLOv8 detects vehicles and pedestrians with ≥90% accuracy
- [ ] DeepSORT tracks objects across frames with ≥85% ID consistency
- [ ] Anomaly detection identifies accidents with ≥90% precision, ≥85% recall
- [ ] GPS coordinates assigned to detected accidents
- [ ] SMS alerts delivered to registered contacts within 5 seconds
- [ ] Email alerts delivered with full incident details
- [ ] Web dashboard displays detection results and alerts
- [ ] Alert history logged and exportable to CSV

Performance Acceptance:

- [ ] Processing speed: ≥25 FPS on GPU, ≥5 FPS on CPU
- [ ] Alert delivery time: <5 seconds (95th percentile)
- [ ] Alert delivery success rate: >99%
- [ ] System uptime: ≥99% during testing period
- [ ] Memory usage: <1 GB during processing

Usability Acceptance:

- [ ] System usable by operators after 4 hours of training
- [ ] User satisfaction score: ≥4.0/5.0
- [ ] No critical usability issues identified in testing
- [ ] Documentation complete and understandable

10.2 Production Deployment Criteria

System Readiness:

- [ ] All MVP acceptance criteria met
- [ ] Live stream processing operational
- [ ] Multi-camera support (5+ cameras) tested
- [ ] User authentication and role management working
- [ ] Production database (PostgreSQL) deployed
- [ ] Monitoring and alerting configured
- [ ] Backup and disaster recovery procedures documented

Stakeholder Readiness:

- [ ] Government approval obtained
- [ ] Emergency services integrated and trained
- [ ] System operators trained (at least 3 personnel)
- [ ] Legal and privacy compliance verified
- [ ] Pilot program results demonstrate effectiveness
- [ ] Funding secured for 1-year operation

Operational Readiness:

- [ ] 24/7 support plan established
- [ ] Incident response procedures documented
- [ ] Performance monitoring dashboard operational
- [ ] System maintenance schedule defined
- [ ] User documentation and training materials complete

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11. Appendices

11.1 Glossary

- **CCTV:** Closed-Circuit Television - video surveillance system
- **YOLOv8:** You Only Look Once version 8 - real-time object detection model
- **DeepSORT:** Deep Simple Online and Realtime Tracking algorithm
- **GPS:** Global Positioning System - satellite-based navigation
- **Precision:** True positives / (True positives + False positives) - accuracy of positive predictions
- **Recall:** True positives / (True positives + False negatives) - coverage of actual positives
- **F1-Score:** Harmonic mean of precision and recall - balanced performance metric
- **Confidence Score:** Model's certainty about detection (0.0-1.0 scale)
- **Latency:** Time delay from input to output
- **ROI:** Return on Investment - financial benefit vs. cost

11.2 References

[1] World Health Organization. (2023). *Global status report on road safety 2023*.

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[3] Karim, F., et al. (2024). Real-time accident detection using YOLOv8 and Deep SORT. *Journal of Intelligent Transportation Systems*.

[4] Prasad, S. L., et al. (2025). AI-powered emergency response systems: Design, implementation, and evaluation. *IEEE Transactions on Intelligent Transportation Systems*.

[5] Ngekeng, A. S., et al. (2024). Road traffic safety in Cameroon: A systematic review. *Safety Science*, 168, 106309.

11.3 Document Revision History

Version	Date	Author	Changes
1.0	Feb 5, 2026	Sangbong Atungsiri Awah	Initial PRD creation - complete product requirements document

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END OF PRODUCT REQUIREMENTS DOCUMENT